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GResMark: A swin transformer-based watermarking framework with geometric attack resilience

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ABSTRACT

Deep learning-based digital watermarking technology plays a crucial role in copyright protection and has been extensively researched. Current CNN-based robust watermarking methods embed the watermark into the deep features of the image through convolution operations. Despite the success of CNN-based watermarking frameworks, their reliance on capturing features at fixed spatial positions makes it challenging to address desynchronization caused by geometric distortions. To solve the above problem, we propose a robust watermarking framework based on the Swin Transformer named GResMark. Intuitively, GResMark leverages the spatial self-attention mechanism of the Swin Transformer to capture global and long-range dependencies among image tokens. Since geometric attacks preserve these relationships, GResMark achieves improved adaptability and robustness against such distortions. Specifically, we design the Locally-enhanced Channel Attention (LeCA) Swin Transformer block and Frequency Channel Attention (FCA) Transformer block as the backbone of our framework, enabling more effective dependency modeling for robust watermarking. In addition, to mitigate the impact of invalid pixel values in the watermarked images generated during training, which exceed the valid range defined by the underlying data format, we introduce a constrained loss that enhances the quality of watermarked images. Experimental results demonstrate that GResMark outperforms existing state-of-the-art (SOTA) watermarking methods in terms of robustness to geometric distortions and watermark capacity. For geometric attacks, GResMark achieves an extraction accuracy exceeding 98 % while maintaining superior visual quality.

1. Introduction

Robust image watermarking is a widely used technique for copyright protection (Chang & Shen, 2017) and serves as a key enabler for image encryption (Hu et al., 2024, 2025; Li et al., 2025b), tamper detection (Zou et al., 2025), and trustworthy AI content management (Li et al., 2025a; Liu et al., 2024; Wu et al., 2021). This technique aims to embed the invisible watermarks into images and extract the watermarks even if the images experience severe distortions. In general, a robust image watermarking technique must possess three key characteristics: robustness, imperceptibility, and watermark capacity. Robustness ensures that the embedded watermark can resist a variety of distortions while maintaining its integrity. Imperceptibility ensures that the watermark is embedded without compromising the original visual quality of the image. Watermark capacity refers to the amount of information that can be embedded without affecting either robustness or imperceptibility. The initial research (van Schyndel et al., 1994) embedded the watermark message into the spatial domain, but this approach is easily detectable (Dumitrescu et al., 2003). In pursuit of robustness, later work focused on embedding watermark messages into the frequency domain (Chen et al., 2024; Fang et al., 2019; Ko et al., 2020). However, these traditional methods rely heavily on hand-crafted shallow image features, which have many limitations in terms of robustness (Jia et al., 2021).

In recent years, with the rapid development of deep learning technology, various deep learning-based watermarking frameworks (Ahmadi et al., 2020; Fang et al., 2023a, 2022a; Fernandez et al., 2022; Guo et al., 2024, 2023; Liu et al., 2023, 2022; Ma et al., 2022; Qin et al.,

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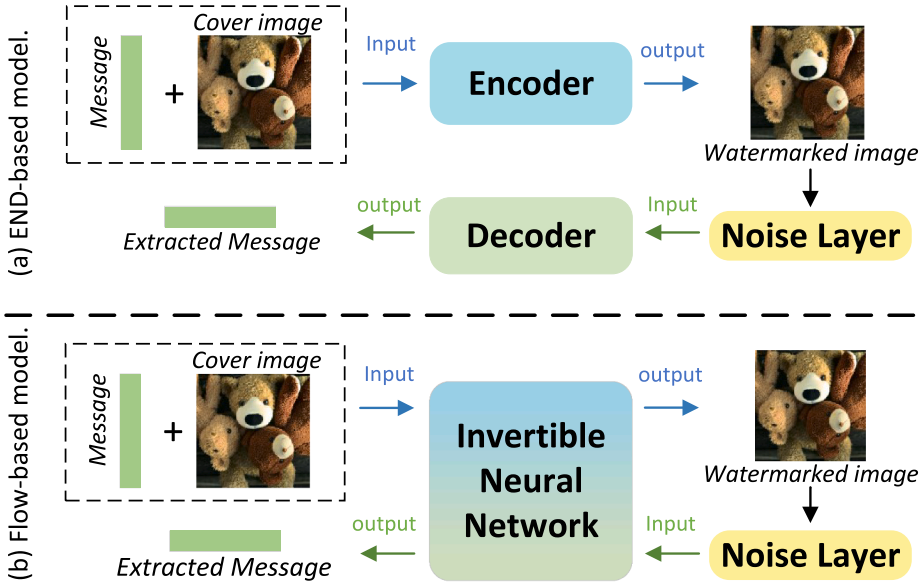


Fig. 1. The difference between END-based model and flow-based model.

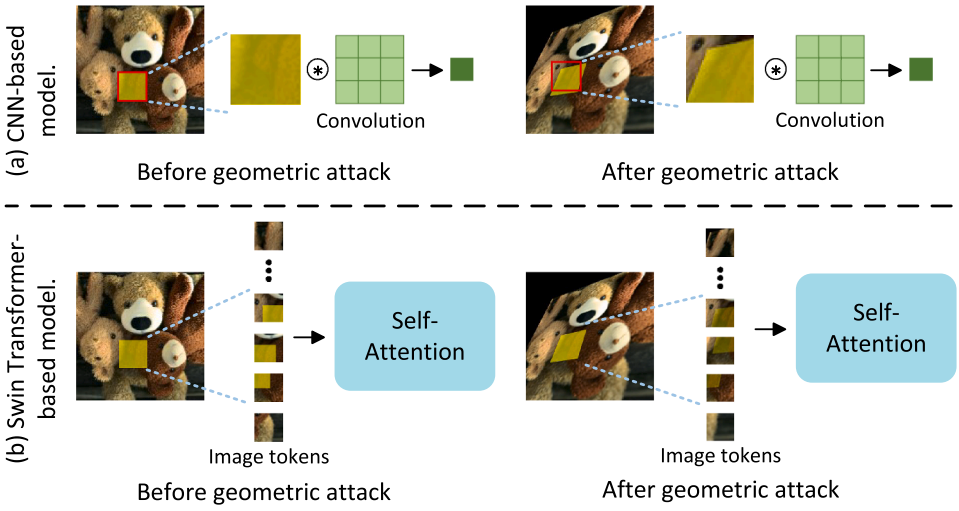


Fig. 2. The difference of watermark feature extraction before and after geometric attacks between the CNN-based model and our Swin Transformer-based model. Yellow regions highlight the key regions containing watermark features, and the red boxes denote the convolutional receptive fields.

2024; Wang et al., 2024a; Wu et al., 2023; Zhang et al., 2020; Zhu et al., 2018) have emerged. These frameworks aim to address the limitations of traditional methods by fully utilizing the rich features of images. Typically, these frameworks consist of an **E**ncoder, a **N**oise layer, and a **D**ecoder (END), as shown in Fig. 1(a). The purpose of the encoder is to embed the watermark message into the cover image. The noise layer then applies various distortions to the watermarked image. The decoder attempts to extract the watermark message from the distorted image. Recently, researchers have proposed a flow-based robust watermarking framework (Fang et al., 2023c), as shown in Fig. 1(b). This framework exploits the reversibility of Invertible Neural Networks (INNs) (Dinh et al., 2014; Jing et al., 2021) to achieve parameter sharing between the encoding and decoding processes, thus reducing redundant feature embedding and improving both imperceptibility and robustness.

Although these watermarking frameworks achieve good imperceptibility and robustness, current frameworks focus on addressing typical non-geometric attacks (e.g., JPEG compression, Salt & Pepper noise), which primarily reduce the strength of the watermark. In contrast, they neglect common geometric attacks in real-world scenarios (e.g., rotation, affine transformations), which lead to watermark desynchronization issues, where the watermark remains intact but becomes misaligned with the reference grid, making reliable extraction difficult (Hosam, 2019). Most of these frameworks are based on CNNs, which face inherent limitations due to the nature of convolution operations. CNN-based frameworks often struggle to model long-range dependencies (Cao et al., 2022), which limits their ability to capture complex spatial relationships. As shown in Fig. 2(a), before a geometric attack, the convolution kernels align well with the salient regions containing watermark features, enabling effective extraction. However, after a geometric attack, desynchronization shifts these regions, preventing the fixed kernels from aligning with the watermark-embedded regions. This inflexibility limits CNNs to capturing feature relationships at fixed spatial locations, making existing CNN-based frameworks inadequate for addressing feature desynchronization caused by geometric attacks. Recently, Luo et al. (2024) proposed WFormer, the first Transformer-based digital watermarking framework, which enhances watermarking by better fusing watermark and image features. It employs the self-attention mechanism of Transformers to expand the watermark and uses cross-attention to

build communication between the image and the watermark. However, WFormer performs self-attention within fixed-size image patches and relies on channel-wise feature modeling, neglecting explicit positional dependencies. As a result, when geometric attacks introduce desynchronization between the watermark and the cover image, channel-wise attention alone cannot effectively resolve the misalignment, thereby limiting robustness against geometric distortions.

In addition, we observe that current watermarking frameworks are trained and tested under ideal experimental conditions, which often leads to pixel values in the watermarked image exceeding the valid range defined by the underlying data format (e.g., [0,255] for 8-bit images). Such invalid values are inevitably clipped during transmission or storage, resulting in visible artifacts in the watermarked images. This pixel truncation issue ultimately undermines the imperceptibility of the watermarked image.

To address these challenges, we propose a Swin Transformer-based framework for robust image watermarking named GResMark. As illustrated in Fig. 2(b), geometric attacks typically cause desynchronization of the watermark while preserving the relative positional relationships among watermark features. The Swin Transformer partitions an image into equally sized patches (image tokens) and applies shifted window self-attention. Unlike CNNs that rely on fixed local receptive fields, the self-attention mechanism in the Swin Transformer computes pairwise correlations among all tokens, enabling the network to capture global contextual dependencies and dynamically model feature relationships across distant regions. This allows the model to retain the watermark features even under geometric attacks. Building on this observation, we design two customized components, the Local-enhanced Channel Attention (LeCA) Swin Transformer block and the Frequency Channel Attention (FCA) Transformer block, as GResMark's backbone to effectively model positional dependencies among tokens. The self-attention mechanism captures the dependencies among tokens, thereby mitigating the desynchronization issue caused by geometric attacks. Meanwhile, LeCA enhances local channel feature extraction, and FCA captures frequency-domain information, jointly improving the robustness of the watermark representation. Additionally, to improve the imperceptibility in real-world scenarios, we introduce a constrained loss that regularizes the encoder to prevent the generation of invalid pixel values, reducing artifacts. Extensive experimental results demonstrate that GResMark is capable of withstanding geometric attacks while maintaining a high quality of the encoded image.

The key contribution of our work can be summarised as follows:

(1) We analyze the fundamental differences between CNNs and Transformers in capturing watermark features under geometric attacks. Based on this insight, we propose GResMark, a Swin Transformer-based robust watermarking framework that effectively models global spatial dependencies to mitigate watermark desynchronization.

(2) We introduce two novel modules, the Locally-enhanced Channel Attention (LeCA) block and the Frequency Channel Attention (FCA) block, which enhance local spatial feature extraction and frequency-domain robustness, respectively, improving watermark imperceptibility and robustness.

(3) We design a constrained loss to prevent the generation of invalid pixel values, thereby reducing artifacts in watermarked images.

The remainder of this paper is organized as follows. Section II introduces the related work of deep learning-based Watermarking and Vision Transformer. Section III introduces the proposed watermarking model based on the Swin Transformer. Section IV evaluates and analyzes the results of the experiment. Section V concludes this paper.

2. Related work

2.1. Deep learning-based watermarking

In recent years, many deep learning-based robust watermarking frameworks have been presented. Zhu et al. (2018) first proposed the END framework, HiDDeN, which successfully achieved robustness against image processing attacks (e.g., JPEG compression, Blurring). Inspired by HiDDeN (Zhu et al., 2018), Tancik et al. (2020) designed StegaStamp, which addresses the challenges of watermarking in print-and-capture scenarios by using a noise layer of differentiable image perturbations that simulates the print-capture process during training. To overcome the limitation of requiring differentiable noise, Liu et al. (2019) introduced a two-stage separable deep learning network (TSDL). By fine-tuning the decoder in the second stage using only real attacks, TSDL enhances robustness against non-differentiable and black-box attacks. In order to improve robustness, Fang et al. (2022b) presented an encoded feature enhancement watermarking network based on TSDL (Liu et al., 2019). Nevertheless, this multi-stage training watermarking framework still lacks robustness against JPEG compression. To enhance robustness against JPEG compression, Jia et al. (2021) proposed a method called MBRS, which achieves high robustness by using mini-batches of "real JPEG" and "simulated JPEG" noise during network training. In contrast, the above END watermarking frameworks suffer from redundant feature embedding by the encoder. To solve this problem, Fang et al. (2023b) proposed De-END that strengthens the coupling between the encoder and decoder by using the decoder to guide the encoder. Subsequently, Fang et al. (2023c) introduced a flow-based watermarking framework that leverages the invertibility of INN to enable weight sharing between the encoder and decoder, ensuring a tightly coupled relationship. Although these methods have made significant progress, they still have shortcomings in the feature extraction and interaction between watermarks and image features. To address these issues, Luo et al. (2024) recently proposed a Transformer-based model called WFormer, which achieves watermark expansion and feature fusion between the image and watermark through the Transformer attention mechanism. However, due to the inherent limitations of CNNs and the fixed-size block with channel-based self-attention used in WFormer, those approaches still face challenges in resisting watermark desynchronization caused by geometric attacks.

2.2. Watermarking resists geometric attacks

Geometric attacks are common and challenging distortions in real-world watermarking scenarios, as they cause watermark desynchronization. Existing studies mainly address this issue by exploiting image features or embedded watermark features to achieve watermark resynchronization after geometric transformations. Hu and Xiang (2021) proposed a robust watermarking algorithm based on low-order Zernike moments, which demonstrates strong robustness against rotation and scaling attacks. Ma et al. (2021) introduced a symmetrical watermark synchronization scheme that embeds symmetrical watermark patterns to effectively resist local geometric distortions. To further enhance resistance to scaling and cropping attacks, Su et al. (2025) proposed an adaptive watermarking method based on the Hadamard transform, which adaptively corrects image scaling through energy distribution analysis, thereby improving watermark stability. Wu et al. (2025) developed a DFT-based watermarking method that combines polar coordinate mapping and machine-learning-based scaling estimation, enabling accurate watermark extraction under arbitrary scaling and cropping scenarios. Chen et al. (2024) designed a DFT-based synchronized watermark template, which achieves strong geometric robustness while maintaining resistance to screen-shooting attacks. In addition, Wang et al. (2024b) proposed a robust blind watermarking algorithm that integrates stable interest point detection with deep neural networks, significantly improving robustness against hybrid and geometric attacks. However, despite the significant progress made in the field of geometric-robust watermarking, most existing methods still rely on hand-crafted features and lack the ability to model global contextual relationships between image regions. Consequently, their robustness tends to degrade significantly when faced with complex or combined geometric attacks.

2.3. Vision transformer

Vision Transformer (Dosovitskiy et al., 2020) has demonstrated superior performance in many vision tasks, such as image classification (Zhang et al., 2024) and object detection (Chen et al., 2022). ViT divides images into patches (also known as tokens) and utilizes a self-attention mechanism to capture long-range dependencies between these patches. However, the quadratic computational complexity of global self-attention limits its application in high-resolution images. To address this issue, Wang et al. (2021) proposed the Pyramid Vision Transformer (PVT), which employs a progressive shrinking pyramid to reduce computational complexity on large feature maps. Chen et al. (2021) introduced CrossViT, a dual-branch structure that combines image patches of different sizes to generate stronger image features. Liu et al. (2021) developed the Swin Transformer, which applies self-attention to local windows using a window-shifting mechanism, thus avoiding the original quadratic complexity and achieving significant results. Subsequently, these Vision Transformer models have been applied to a variety of image-processing tasks. Wang et al. (2022) designed UFormer, a U-shaped model for efficient image restoration. Ke et al. (2024) introduced StegFormer, first applying the Transformer to image steganography. Luo et al. (2024) proposed WFormer, the first digital watermarking framework based on Transformer. Inspired by those works, we propose GResMark, a model based on Swin Transformer specifically designed for image watermarking.

3. Our method

3.1. Motivation

Geometric attacks cause desynchronization of watermarks. Existing CNN-based approaches, such as HiDDeN (Zhu et al., 2018) and TSDL (Liu et al., 2019), focus primarily on embedding the watermark within deep features using convolutional operations. However, these models rely on fixed receptive fields, making it difficult for them to handle desynchronization of watermarks.

We observe that while geometric attacks introduce positional misalignments, they do not alter the relative relationships among watermark features. If these relationships can be effectively modeled globally, the desynchronization caused by geometric attacks can be mitigated. This insight motivates our approach to leverage the self-attention mechanism in Vision Transformers (Dosovitskiy et al., 2020). Unlike CNNs, Vision Transformers model the relationships between all tokens (image patches), regardless of their spatial distance, making them highly effective at preserving watermark features even after severe geometric attacks.

Based on this motivation, we propose a robust watermarking framework based on Swin Transformer, named GResMark, as shown in Fig. 3 and Algorithm 1. The entire framework consists of three main components: an encoder E with parameters θ_E, a noise layer N, and a decoder D with parameters θ_D. The cover image I_{co} and the watermark message M_{en} are first fed into E to generate the watermarked image I_{em}. Then, the noise layer N applies different attacks to I_{em}, generating the noise image I_{no}. Lastly, D attempts to extract the watermark message M_{en} from I_{no}. In the following sections, we provide a detailed description of the model architecture, LeCA Swin Transformer block, FCA Transformer block, and Constrained Loss.

3.2. Model architecture

Encoder: The primary purpose of E is to embed M_{en} into I_{co} while maintaining the visual quality of I_{co}. In our proposed scheme, E adopts a U-Net like architecture (Ronneberger et al., 2015) with skip connections and multi-scale features to capture both the global structure and the fine details, reducing the distortion caused during the watermark embedding process, as illustrated in Fig. 3. To be specific, given an image $I_{co} \in \mathbb{R}^{3\times H\times W}$, a 3×3 convolutional layer first extracts its low-level features, generating an output $I_{input} \in \mathbb{R}^{C\times H\times W}$. Next, we divide $I_{input} \in \mathbb{R}^{C\times H\times W}$ into non-overlapping patches of size $P\times P$ and reshape them into a flattened 2D patch sequence $X_{token} \in \mathbb{R}^{\frac{HW}{P^2}\times P^2C}$. Here, (H, W) are the dimensions of I_{co}, C is the number of channels, (P, P) is the size of each patch, and $N = \frac{HW}{P^2}$ represents the total number of patches (or tokens). In the down-convolution part, X_{token} process through K LeCA Swin Transformer blocks and downsampling layers to extract the feature of I_{co}. Each downsampling layer utilises a 4×4 convolutional layer with a stride of 2, which effectively doubles the number of channels while halving the resolution of the feature maps.

At the bottleneck layer of E, an FCA Transformer block is applied to further capture global dependencies from the deep features. By introducing channel attention in the frequency domain, the FCA Transformer block enhances the robustness of the model.

Meanwhile, $M_{en} \in \mathbb{R}^L$, a vector of length L, is processed through several diffusion blocks and upsample layers to expand the watermark features. Specifically, M_{en} first passes through a linear layer, generating an output vector of length L_1, which is then reshaped into a matrix of size $L_2\times L_2$. The nearest-neighbor interpolation method resizes M_{en} to match the dimensions of the corresponding feature maps of I_{co} from the downsampling stage. Finally, a 3×3 convolutional layer increases the number of feature channels to C_1.

In the up-convolution part, K LeCA Swin Transformer blocks and upsampling layers are used to fuse the feature of I_{co} and M_{en}. Each upsampling layer employs a 2×2 transposed convolution with a stride of 2, which halves the number of channels and doubles the spatial dimensions of the feature map. The upsampled feature map is then concatenated with the symmetry feature map and the expanded watermark feature map. This concatenated feature map is subsequently fed into the LeCA Swin Transformer block for feature fusing. Ultimately, we obtain $X_{output} \in \mathbb{R}^{\frac{HW}{P^2}\times P^2(2C+C_1)}$, which is reshaped into an image $I_{output} \in \mathbb{R}^{(2C+C_1)\times H\times W}$. A 3×3 convolutional layer is applied to reduce the dimensionality of I_{output}, producing the final watermarked image $I_{em} \in \mathbb{R}^{3\times H\times W}$.

The aim of E is generate I_{em} which approach I_{co} by updating θ_E, following the loss L_E:

$$\begin{aligned} L_E &= \mathrm{MSE}(I_{co}, I_{em}) \\ &= \mathrm{MSE}(I_{co}, E(\theta_E, I_{co}, M_{en})), \end{aligned} \tag{1}$$

where $\mathrm{MSE}(\cdot)$ computes the mean square error.

Noise Layer: The incorporation of a noise layer N during training plays a crucial role in enhancing the robustness of the watermark (Hayes & Danezis, 2017). We employ various differentiable image processing operations as the noise layer N in our GResMark, distorting the watermarked image I_{em} into a noise image I_{no}. Since real JPEG compression is non-differentiable, we use an existing differentiable noise layer (Shin & Song, 2017) to simulate the JPEG compression process.

Decoder: The primary purpose of D is to extract the watermark message from I_{no}. We construct D by referring to the down-convolution part of the U-net, comprising K LeCA Swin Transformer blocks, downsampling layers, an FCA Transformer block, and an information extraction layer. First, We apply a 3×3 convolution to the noise image $I_{no} \in \mathbb{R}^{3\times H\times W}$ to extract shallow features, resulting in $I^{de}_{input} \in \mathbb{R}^{C\times H\times W}$. Then, I^{de}_{input} is divided and reshaped into a sequence $X^{de}_{token} \in \mathbb{R}^{\frac{HW}{P^2}\times P^2C}$. By passing through K LeCA Swin Transformer blocks, K downsampling layers and one FCA Transformer block, we obtain the watermark feature map $I^{output}_{no} \in \mathbb{R}^{2^K C\times \frac{H}{2^K}\times \frac{W}{2^K}}$. Finally, I^{output}_{no} is fed into the information extraction layer, which consists of a convolutional layer and a fully connected layer, to extract the watermark message $M_{ex} \in \mathbb{R}^L$.

The aim of D is generate M_{ex} which approach the original watermark message M_{en} by updating θ_D, which can be formulated by:

$$L_D = \mathrm{MSE}(M_{en}, M_{ex})$$

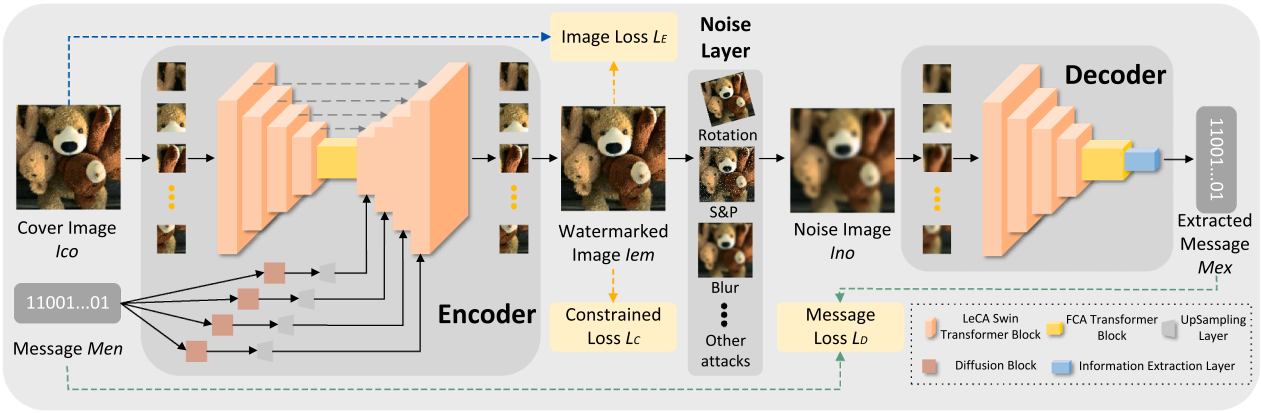


Fig. 3. The framework of GResMark. The encoder E and decoder D consist of two crucial components: the LeCA Swin Transformer block and the FCA Transformer block. The encoder E takes the cover image I_{co} and watermark message M_{en} as input and produces the watermarked image I_{em} as output. The decoder D receives the noise image I_{no} as input and outputs the extracted watermark message M_{ex}. Between the encoder and decoder is a noise layer N, which includes both non-geometric and geometric attacks.

Algorithm 1 GResMark training framework.

Input: training cover images I_{co}, watermark message M_{en}, watermark encoder E, noise layer N, watermark decoder D, number of iterations $maxiter$.
Output: best θ_E and θ_D.

```
Initialization;
for i ∈ [0, maxiter] do
  for I_co ∈ each batch b do
    // encoding
    I_em ← E(I_co, M_en);
    Compute image loss L_E with Eq. (1);
    Compute constrained loss L_C with Eq. (5);
    // noising
    I_no ← N(I_em);
    // decoding
    M_ex ← D(I_no);
    Compute message loss L_D with Eq. (2);
    L_total = λ1 L_E + λ2 L_D + λ3 L_C;
    Update θ_E and θ_D;
  end for
end for
Return: best θ_E and θ_D.
```

$$= \mathrm{MSE}(M_{en}, D(\theta_D, I_{no})). \tag{2}$$

3.3. LeCA Swin Transformer block

Applying Vision Transformer (Dosovitskiy et al., 2020) to image watermarking tasks presents three main challenges. (1) The quadratic computational cost of the self-attention mechanism renders Vision Transformers inefficient for handling complex visual tasks. (2) Previous studies (Li et al., 2023, 2021) have shown that Vision Transformer struggles to capture local and channel features, both of which are essential for robust image watermarking. (3) Cross-channel covariance is calculated instead of spatial dimensions, which lacks the capture of spatial dimension information, resulting in insufficient performance against geometric attacks.

To address these challenges, we propose the LeCA Swin Transformer block as a fundamental component of GResMark, as illustrated in Fig. 4. Specifically, the LeCA Swin Transformer block consists of two main parts: the first part is Swin Transformer block (Liu et al., 2021), which employs a window-based self-attention mechanism to effectively reduce computational costs, making the Transformer more suitable for image

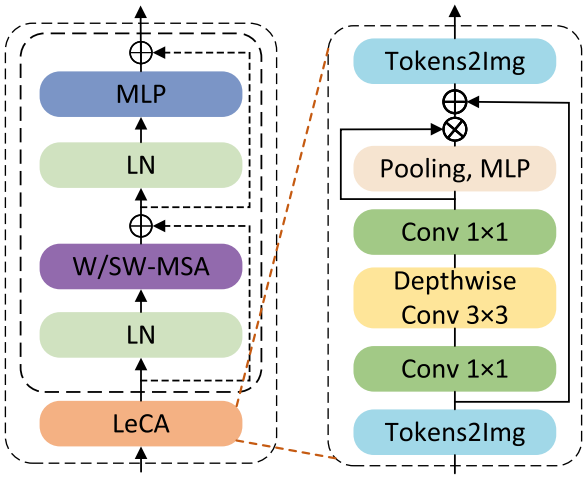


Fig. 4. Illustration of LeCA Swin Transformer block. This block consists of LeCA and the standard Swin Transformer block. LeCA enhances the feature extraction ability of Swin Transformer by using depthwise convolution and channel attention to capture both local and global features.

watermarking tasks. However, the window-based self-attention mechanism has limitations in capturing the channel features of images. To overcome this, we introduce the second part: LeCA. LeCA introduces depthwise convolution to emphasize local context and employs channel attention to enhance GResMark's ability to capture channel features. Unlike existing attention enhancement modules, LeCA is specifically designed for the Swin Transformer to strengthen spatial-channel feature coupling in watermark embedding, thereby ensuring more stable feature preservation under geometric distortions.

Swin Transformer Block: Unlike the Vision Transformer (Dosovitskiy et al., 2020), which computes self-attention over the entire image, Swin Transformer (Liu et al., 2021) computes self-attention within independent windows, significantly reducing computational costs. The Swin Transformer block consists of a LayerNorm (LN) layer, window-based multi-head self-attention (W-MSA), residual connections, and a 2-layer MLP with GELU nonlinearity. Due to the lack of inter-window self-attention in W-MSA, Swin Transformer uses two consecutive Swin Transformer blocks and a shifted window approach to establish connections between windows. This consecutive Swin Transformer block (Liu et al., 2021) with shifted windows can be represented as:

$$\hat{X}^l = \text{W-MSA}(\text{LN}(X^{l-1})) + X^{l-1},$$

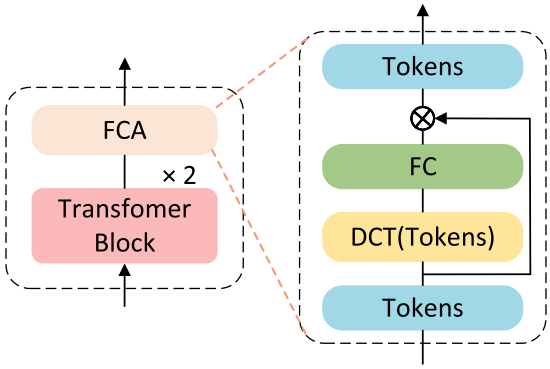


Fig. 5. Illustration of FCA Transformer block. This block combines FCA and the standard Transformer block. It leverages Vision Transformer for global feature extraction and applies FCA's frequency-domain attention to improve robustness against distortions.

$$
\begin{aligned}
X^l &= \mathrm{MLP}(\mathrm{LN}(\hat{X}^l)) + \hat{X}^l, \\
\hat{X}^{l+1} &= \text{SW-MSA}(\mathrm{LN}(X^l)) + X^l, \\
X^{l+1} &= \mathrm{MLP}(\mathrm{LN}(\hat{X}^{l+1})) + \hat{X}^{l+1},
\end{aligned} \tag{3}
$$

where $\hat{X}^l$ and X^l denote the output features of the (S)W-MSA module and the MLP module for block l, respectively.

Locally-enhanced Channel Attention (LeCA): Transformers focus on global modeling, which limits their ability to capture local features and channel information. To improve the performance of GResMark, we propose the LeCA to extract both local features and channel information from images effectively. As shown in Fig. 4, we first reshape $X_{token} \in \mathbb{R}^{\frac{HW}{P^2} \times P^2 C}$ into an image $X_{img} \in \mathbb{R}^{C \times H \times W}$. A 1×1 convolution is applied to increase the dimensionality, allowing better capture of channel information, followed by a 3×3 depthwise convolution to capture local features. Afterwards, another 1×1 convolution is applied to reduce the number of channels, aligning the output with the input dimensions and yielding $X_{channel} \in \mathbb{R}^{C \times H \times W}$. Next, a pooling layer followed by an MLP layer computes attention weights for each channel, resulting in $X_{weight} \in \mathbb{R}^C$. We multiply $X_{channel}$ by X_{weight} to generate a bias $X_{bias} \in \mathbb{R}^{C \times H \times W}$ that incorporates both local features and channel information. Finally, we add X_{bias} to X_{img}, reshape the result back into tokens, and use it as input for the Swin Transformer block.

3.4. FCA Transformer block

Although the Swin Transformer (Liu et al., 2021) employs a shifted window approach to establish connections between windows, its window-based self-attention still limits the ability to capture global features. Given that the image undergoes multiple downsampling operations, the size of the deep feature map is significantly reduced. As a result, even with global self-attention, the computational cost remains relatively low. Therefore, in the FCA Transformer block, we use the Vision Transformer instead of Swin Transformer to further capture the global dependencies of the deep feature map.

Moreover, network models often struggle to capture rich frequency information. Prior work (Cao et al., 2024) has shown that the frequency changes cause image distortion and compromise the embedded watermark, affecting its integrity and robustness. To address this issue, we introduce FCA to guide the model in learning information from the frequency-domain of the deep feature map, as shown in Fig. 5. Specifically, the FCA Transformer block is composed of two Vision Transformer blocks (Dosovitskiy et al., 2020) and the FCA. Different from conventional Transformer modules, FCA adaptively models the importance of frequency components, enhancing the model's resilience to frequency-domain distortions such as blurring.

Transformer Block: We employ Transformer block (Dosovitskiy et al., 2020) to effectively capture global features across the entire image. The computation of the Transformer block is as follows:

$$
\begin{aligned}
\hat{Y}^l &= \mathrm{MSA}(\mathrm{LN}(Y^{l-1})) + Y^{l-1}, \\
Y^l &= \mathrm{MLP}(\mathrm{LN}(\hat{Y}^l)) + \hat{Y}^l,
\end{aligned} \tag{4}
$$

where $\hat{Y}^l$ and Y^l denote the output features of the MSA module and the MLP module for block l, respectively.

Frequency Channel Attention (FCA): We designed the FAC, inspired by FcaNet (Qin et al., 2021), to extract frequency domain features from image tokens, as shown in Fig. 5. First, we divide the Transformer output $X^T_{token} \in \mathbb{R}^{\frac{HW}{P^2} \times P^2 C}$ into n groups along the channel dimension. Each group is regarded as a spatial feature map, to which a corresponding 2D DCT frequency component is applied to transform the features into the frequency domain. Subsequently, the frequency components of all groups are compressed and concatenated into the frequency domain attention vector $X_{\text{Freq}} \in \mathbb{R}^{\frac{HW}{P^2} \times P^2 C}$. Finally, a lightweight fully connected (FC) layer produces channel-wise frequency attention weights $F_w \in \mathbb{R}^{P^2 C}$. The output of the Frequency Enhance block X_{output} is obtained by multiplying F_w with X^T_{token}.

3.5. Constrained loss

During network training, all images, including I_{co} and I_{em}, are normalized to the range $[0, 1]$ to ensure numerical stability and facilitate loss computation. However, during watermark embedding, some pixel values in I_{co} fall outside the standard range of $[0, 1]$. To address this issue, we propose a constrained loss function L_C, which encourages E to ensure that the pixel values of I_{em} remain within the $[0, 1]$ range, improving the imperceptibility of the watermark. L_C is defined as follows:

$$
L_C = \sum_{i=1}^{H} \sum_{j=1}^{W} \begin{cases} 1/2|I_{em}(i,j) - 1|, & \text{if } I_{em}(i,j) > 1, \\ 1/2|I_{em}(i,j)|, & \text{if } I_{em}(i,j) < 0, \\ 0, & \text{otherwise.} \end{cases} \tag{5}
$$

Therefore, the total loss function L_{total} consists of image loss, message loss, and constrained loss, which can be formulated by:

$$
L_{total} = \lambda_1 L_E + \lambda_2 L_D + \lambda_3 L_C, \tag{6}
$$

where λ_1, λ_2, and λ_3 are weight factors.

4. Experimental evaluation

4.1. Experimental settings

Basic Settings: Our GResMark model is implemented using PyTorch (Collobert et al., 2011) and executed on an NVIDIA GeForce RTX 4090 GPU. We use the AdamW optimizer to train GResMark, applying a cosine decay strategy to gradually reduce the learning rate from an initial value of $2e-4$ to $1e-6$. The batch size is set to 16, and the number of epochs is set to $3K$. The weight factors for the loss function are defined as $\lambda_1 = 4$, $\lambda_2 = 10$, and $\lambda_3 = 0.001$. The channel number C for low-level features is set to 16, the watermark extension length L_1 is 256, and the number K for LeCA Swin Transformer blocks is set to 4.

Datasets: We train the model using the DIV2K (Agustsson & Timofte, 2017) dataset. For testing, we use the DIV2K validation set together with $2,000$ images randomly selected from COCO (Lin et al., 2014) and ImageNet (Deng et al., 2009).

Benchmarks: To evaluate the imperceptibility, robustness, and watermark capacity of the proposed GResMark, we compare it with several SOTA watermarking methods, including two classical methods ((Wu et al., 2025) and RtScW (Chen et al., 2024)); four CNN-based END methods (HiDDeN (Zhu et al., 2018), TSDL (Liu et al., 2019), MBRS (Jia et al., 2021), and SepMark (Wu et al., 2023)); a flow-based normalization method (FIN (Fang et al., 2023c)); and a Transformer-based method

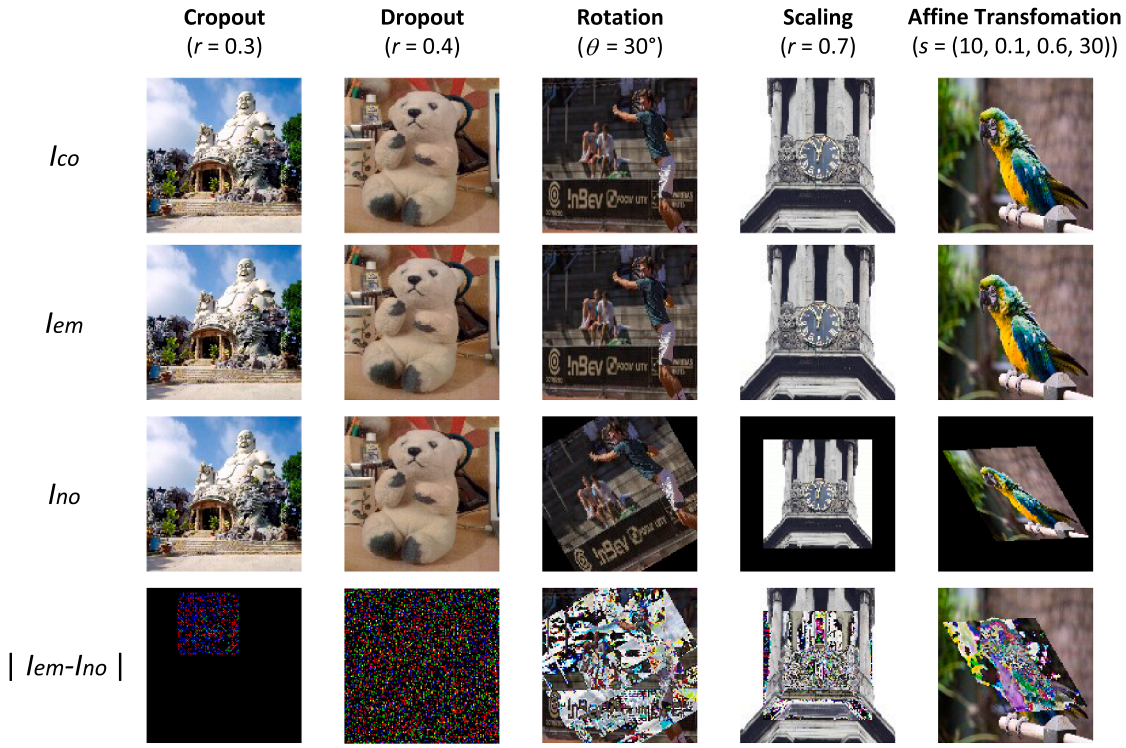


Fig. 6. Watermarked images and corresponding images with geometric distortions. Top: the cover image I_{co}; Second: the encoded image I_{em}; Third: the noise image I_{no}; Bottom: the residual image $|I_{em} - I_{no}|$.

(WFormer (Luo et al., 2024)). All SOTA methods are implemented under identical experimental settings to ensure a fair comparison. To ensure consistency between methods, all images are resized to 128×128. Unless otherwise specified, the default watermark length L is set to 64.

Metrics: In this paper, the Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index Measure (SSIM) are used to evaluate the imperceptibility of the watermark. Meanwhile, the extraction bit accuracy (ACC) evaluates the robustness.

4.2. Imperceptibility and robustness against specific geometric distortion training

In this section, we evaluate the visual quality and robustness of our method against different types of distortions. Specifically, we consider five commonly applied geometric distortions: cropout, dropout, rotation, scaling, and affine transformation, as illustrated in Fig. 6. For each type of distortion, we trained a specialized network to assess the model's performance under different attack conditions. As shown in Fig. 7, the PSNR and SSIM values of our proposed method are compared with those of the benchmark methods under various geometric distortions.

Cropout: Cropout is a type of distortion where part of the watermarked image is preserved, while the remaining area is replaced with the corresponding region from the original image. During the training stage, a cropout ratio of 0.4 is applied. In the testing phase, we evaluate the performance of different models by varying the cropout ratio from 0.1 to 0.5. The experimental results are presented in Fig. 7 and Table 1.

As shown in Fig. 7, the proposed method achieves a remarkable PSNR of 54.74 dB and an SSIM of 0.9994 for watermarked images, ensuring superior visual quality. Despite maintaining this high level of image quality, the method also demonstrates outstanding robustness, with extraction accuracy consistently exceeding 99 % across cropout ratios ranging from 0.1 to 0.5. Notably, as the cropout ratio increases, the advantages of the proposed model become more pronounced, further highlighting its effectiveness.

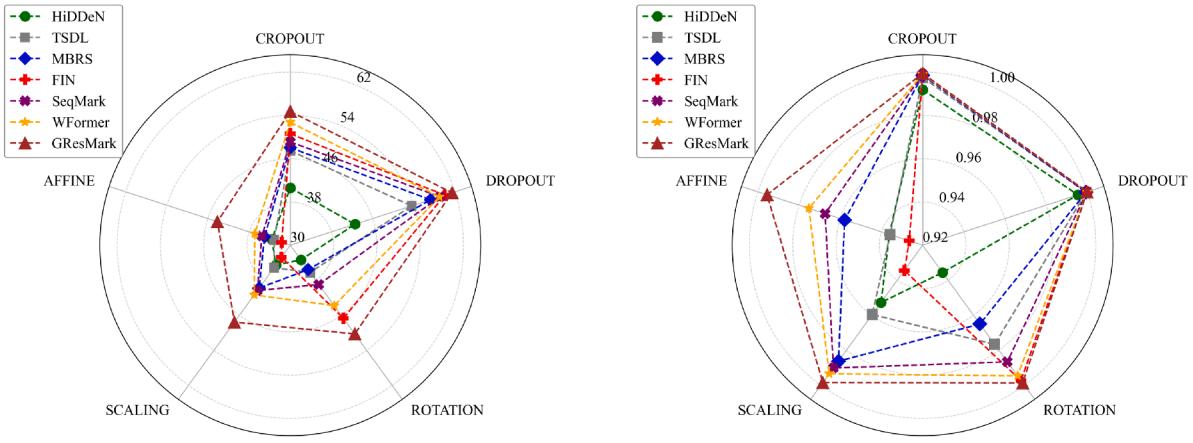
Table 1
Extraction accuracy with different ratios of cropout.

Models	Ratio				
	0.1	0.2	0.3	0.4	0.5
HiDDeN (Zhu et al., 2018)	95.63 %	94.73 %	88.75 %	76.88 %	61.67 %
TSDL (Liu et al., 2019)	98.72 %	98.54 %	96.88 %	93.75 %	93.21 %
MBRS (Jia et al., 2021)	99.71 %	99.22 %	97.18 %	90.43 %	83.50 %
FIN (Fang et al., 2023c)	**100 %**	99.34 %	97.90 %	96.55 %	93.82 %
SepMark (Wu et al., 2023)	99.94 %	99.29 %	97.76 %	96.02 %	93.37 %
WFormer (Luo et al., 2024)	99.99 %	99.98 %	99.97 %	99.96 %	98.35 %
GResMark	**100 %**	**100 %**	**99.99 %**	**99.98 %**	**99.23 %**

Table 2
Extraction accuracy with different ratios of dropout.

Models	Ratio				
	0.2	0.3	0.4	0.5	0.6
HiDDeN (Zhu et al., 2018)	90.21 %	89.58 %	87.08 %	86.74 %	82.71 %
TSDL (Liu et al., 2019)	97.54 %	95.21 %	93.54 %	92.29 %	90.42 %
MBRS (Jia et al., 2021)	96.29 %	94.73 %	94.15 %	92.58 %	90.63 %
FIN (Fang et al., 2023c)	**100 %**	**100 %**	**100 %**	99.74 %	98.86 %
SepMark (Wu et al., 2023)	**100 %**	99.53 %	99.06 %	98.51 %	97.67 %
WFormer (Luo et al., 2024)	99.59 %	99.22 %	98.68 %	97.66 %	95.84 %
GResMark	**100 %**	**100 %**	**100 %**	**99.98 %**	**99.16 %**

Dropout: Dropout involves the random replacement of a certain percentage of pixels in the watermarked image with pixels from the original image, differing from cropout, where the replacement occurs in a specific region. In dropout, the replacement pixels are randomly distributed across the entire image. During training, a dropout ratio of 0.4 is applied. During testing, this ratio is varied from 0.2 to 0.6 to evaluate performance under different levels of distortion, as detailed in Fig. 7 and Table 2.



(a) PSNR of watermarked images with different methods.

(b) SSIM of watermarked images with different methods.

Fig. 7. Visual quality of watermarked images with geometric distortions.

Table 3
Extraction accuracy with different angle of rotation.

Models	Angle						
	1°	30°	60°	90°	120°	150°	180°
HiDDeN (Zhu et al., 2018)	95.34 %	85.14 %	85.48 %	86.24 %	85.58 %	85.12 %	87.15 %
TSDL (Liu et al., 2019)	95.71 %	81.99 %	81.43 %	82.69 %	82.15 %	81.51 %	82.41 %
MBRS (Jia et al., 2021)	94.88 %	79.75 %	79.72 %	80.85 %	79.35 %	78.62 %	79.89 %
FIN (Fang et al., 2023c)	94.27 %	52.25 %	49.93 %	49.61 %	50.18 %	49.73 %	50.54 %
SepMark (Wu et al., 2023)	96.37 %	88.46 %	87.86 %	92.92 %	88.72 %	87.53 %	92.14 %
WFormer (Luo et al., 2024)	**100 %**	92.63 %	90.54 %	91.92 %	90.63 %	90.88 %	91.91 %
GResMark	**100 %**	**100 %**	**99.99 %**	**100 %**	**99.98 %**	**99.97 %**	**100 %**

With respect to imperceptibility, as shown in Fig. 7, the proposed GResMark achieves the highest PSNR and SSIM values among all methods. In terms of robustness, our method achieved an extraction accuracy of over 99.96 % while maintaining the superior image quality.

Rotation: Rotation involves rotating an image by a specific angle, which can significantly disrupt the alignment of embedded watermarks and hinder accurate extraction. During training, rotation angles are randomly selected within the range of −180° to 180°. In the testing phase, we evaluate the model's performance by applying rotation angles within the range of 1° to 180°. The experimental results for rotation distortion are presented in Fig. 7 and Table 3.

Compared with other methods, our proposed GResMark demonstrates exceptional performance under rotational distortions. It achieves a PSNR exceeding 50 dB and an SSIM above 0.998, indicating the superior visual quality of the watermarked images. Additionally, in terms of robustness, the extraction accuracy approaches 100 % across all tested rotation angles, showcasing the model's exceptional ability to withstand rotational attacks. This significant improvement over existing methods highlights GResMark's capacity to maintain watermark integrity under rotation distortion. This robustness can be attributed to the shifted-window self-attention mechanism of the Swin Transformer, which models global spatial dependencies and effectively preserves the relative positional relationships among watermark features even when the image is rotated.

Scaling: Scaling distortion involves resizing the image to different dimensions, potentially affecting the integrity of embedded watermarks. During this process, bilinear interpolation is applied to resample pixel values when enlarging or shrinking the image. To assess the model's resilience to size variations, scaling factors are randomly selected between 0.7 and 1.2 times the original image size during training. In the testing phase, robustness is evaluated using scaling factors ranging from 0.5 to 1.5. The experimental results for this scaling distortion are shown in Fig. 7 and Table 4.

Table 4
Extraction accuracy with different ratios of scaling.

Models	Ratio				
	0.5	0.7	0.9	1.2	1.5
HiDDeN (Zhu et al., 2018)	69.12 %	84.38 %	88.49 %	85.09 %	72.22 %
TSDL (Liu et al., 2019)	61.10 %	75.35 %	79.91 %	76.19 %	62.86 %
MBRS (Jia et al., 2021)	70.60 %	89.15 %	93.72 %	91.98 %	72.83 %
FIN (Fang et al., 2023c)	50.16 %	68.20 %	73.21 %	75.73 %	55.47 %
SepMark (Wu et al., 2023)	71.05 %	90.74 %	96.00 %	92.02 %	74.64 %
WFormer (Luo et al., 2024)	91.29 %	96.89 %	**100 %**	98.07 %	93.47 %
GResMark	**98.76 %**	**99.58 %**	**100 %**	**100 %**	**99.85 %**

As demonstrated in Fig. 7, the proposed GResMark model ensures high-quality watermarked images, achieving a PSNR exceeding 47 dB and an SSIM of 0.9984. Regarding robustness, when the scaling attack ratio is weak (0.9 and 1.2), our GResMark ACC reaches 100 %. As the scaling attack ratio further increases, although our GResMark's accuracy slightly decreases, it consistently remains above 98 %. This fully demonstrates our GResMark's exceptional performance and reliability in coping with scaling attacks. This strong performance is due to GResMark's multi-scale feature extraction capability and Swin Transformer blocks, which capture local features at different resolutions and maintain feature consistency even when the image size changes, effectively mitigating scaling-induced desynchronization.

Affine Transformation: Affine transformation involves manipulating images through rotation, translation, scaling, and shearing transformations, which can significantly distort the embedded watermark and hinder accurate extraction. During training, the affine parameters are

Table 5
Extraction accuracy with different strengths of affine transformation.

Models	Strengths			
	(10, 0.1, 0.7, 30)	(0, 0.2, 0.7, 30)	(0, 0.1, 0.6, 30)	(0, 0.1, 0.7, 20)
HiDDeN (Zhu et al., 2018)	70.03 %	68.72 %	73.19 %	77.88 %
TSDL (Liu et al., 2019)	68.16 %	65.57 %	70.11 %	68.96 %
MBRS (Jia et al., 2021)	88.39 %	81.62 %	84.35 %	92.11 %
FIN (Fang et al., 2023c)	82.29 %	85.49 %	74.32 %	88.81 %
SepMark (Wu et al., 2023)	90.25 %	83.34 %	87.22 %	96.31 %
WFormer (Luo et al., 2024)	91.79 %	89.63 %	93.32 %	**100 %**
GResMark	**99.08 %**	**99.98 %**	**99.15 %**	**100 %**

Table 6
Visual quality with combined distortion training.

Models	(Wu et al., 2025)	RtScW (Chen et al., 2024)	HiDDeN (Zhu et al., 2018)	TSDL (Liu et al., 2019)
PSNR(dB)	36.03	36.07	30.25	32.76
SSIM	0.9567	0.9502	0.9286	0.9422
MBRS (Jia et al., 2021)	FIN (Fang et al., 2023c)	SepMark (Wu et al., 2023)	WFormer (Luo et al., 2024)	GResMark
34.64	34.51	34.87	35.77	**36.15**
0.9388	0.9450	0.9479	0.9652	**0.9813**

Table 7
Extraction accuracy with single distortions.

Attacks	Models								
	(Wu et al., 2025)	RtScW	HiDDeN	TSDL	MBRS	FIN	SepMark	WFormer	GResMark
Identity	**100 %**	**100 %**	97.26 %	**100 %**	**100 %**	**100 %**	**100 %**	**100 %**	**100 %**
JPEG($QF = 50$)	70.03 %	96.84 %	70.55 %	78.13 %	96.45 %	**99.28 %**	97.92 %	98.55 %	98.43 %
JPEG($QF = 60$)	80.24 %	98.36 %	73.66 %	81.56 %	97.53 %	**99.88 %**	98.70 %	99.81 %	99.77 %
JPEG($QF = 70$)	84.45 %	99.77 %	74.24 %	84.04 %	98.57 %	**100 %**	99.49 %	**100 %**	**100 %**
GN($\sigma = 0.03$)	88.87 %	56.01 %	87.92 %	90.97 %	98.05 %	**98.85 %**	95.16 %	98.29 %	98.25 %
GN($\sigma = 0.04$)	87.42 %	54.99 %	85.49 %	88.39 %	96.44 %	**97.57 %**	93.86 %	97.22 %	97.01 %
GN($\sigma = 0.05$)	84.36 %	50.04 %	80.47 %	84.82 %	94.59 %	**96.12 %**	91.33 %	95.44 %	95.95 %
GB($\sigma = 8$)	70.49 %	65.19 %	61.78 %	83.11 %	98.18 %	94.70 %	98.62 %	97.21 %	**100 %**
GB($\sigma = 4$)	75.18 %	77.53 %	75.37 %	90.26 %	99.04 %	96.93 %	99.43 %	98.75 %	**100 %**
GB($\sigma = 1$)	93.80 %	90.84 %	93.45 %	98.84 %	**100 %**	99.15 %	**100 %**	99.53 %	**100 %**
MB($W = 7$)	72.74 %	85.89 %	77.48 %	91.25 %	96.55 %	95.15 %	97.69 %	97.41 %	**100 %**
MB($W = 5$)	79.77 %	90.33 %	82.03 %	96.23 %	99.76 %	97.98 %	99.20 %	**100 %**	**100 %**
MB($W = 3$)	87.35 %	94.61 %	85.71 %	98.33 %	**100 %**	**100 %**	**100 %**	**100 %**	**100 %**
SP($\sigma = 0.08$)	80.54 %	83.75 %	60.32 %	83.73 %	96.27 %	99.31 %	97.50 %	98.75 %	**99.38 %**
SP($\sigma = 0.06$)	86.57 %	87.23 %	71.02 %	86.97 %	98.34 %	99.75 %	98.95 %	99.47 %	**99.94 %**
SP($\sigma = 0.04$)	91.49 %	92.54 %	79.19 %	89.16 %	99.07 %	**100 %**	99.62 %	**100 %**	**100 %**
Scaling($r = 0.6$)	96.12 %	57.32 %	74.16 %	64.54 %	80.49 %	53.29 %	81.12 %	88.53 %	**98.46 %**
Scaling($r = 0.9$)	97.65 %	97.83 %	85.80 %	79.30 %	92.97 %	60.35 %	94.21 %	96.40 %	**100 %**
Scaling($r = 1.1$)	96.41 %	98.05 %	85.03 %	79.41 %	91.32 %	58.24 %	92.59 %	94.66 %	**98.98 %**
Cropping($r = 0.6$)	89.90 %	**100 %**	87.00 %	92.70 %	88.38 %	93.19 %	92.69 %	94.67 %	98.24 %
Cropping($r = 0.4$)	92.85 %	**100 %**	93.63 %	96.91 %	94.76 %	97.84 %	97.50 %	97.95 %	**100 %**
Cropping($r = 0.2$)	95.53 %	**100 %**	96.46 %	98.67 %	98.62 %	99.73 %	99.01 %	99.54 %	**100 %**
Rotation($\theta = 120°$)	65.93 %	52.61 %	75.37 %	71.46 %	70.78 %	49.75 %	80.75 %	88.16 %	**100 %**
Rotation($\theta = 30°$)	70.22 %	50.28 %	76.12 %	72.06 %	70.62 %	51.86 %	83.44 %	89.49 %	**100 %**
Rotation($\theta = 10°$)	82.09 %	97.98 %	87.17 %	86.34 %	84.24 %	49.75 %	90.24 %	94.47 %	**100 %**
Affine ($s = (0, 0.1, 0.8, 20)$)	69.84 %	83.96 %	70.29 %	55.07 %	75.92 %	49.18 %	76.16 %	95.75 %	**100 %**
Affine ($s = (10, 0.2, 0.8, 30)$)	63.99 %	79.80 %	64.06 %	50.18 %	75.38 %	49.23 %	81.57 %	86.60 %	**99.97 %**
Affine ($s = (10, 0.2, 0.8, 35)$)	58.75 %	73.22 %	60.52 %	53.20 %	63.92 %	50.40 %	69.65 %	80.08 %	**95.44 %**

configured as follows: rotation angles randomly selected between −0° and 0°, translations up to 0.1 in both horizontal and vertical directions, scaling factors set to 0.7 (reducing the image size by 30 %), and shearing angles ranging from −30° to 30°. In the testing phase, robustness is assessed by applying four different levels of affine transformation strength, following the methodology outlined in WFormer (Luo et al., 2024). As shown in Fig. 7 and Table 5, we evaluated and compared our proposed GResMark against other SOTA models under these varying affine transformation strengths.

The experimental results demonstrate that GResMark significantly outperforms existing models in terms of both imperceptibility and robustness. Specifically, our GResMark achieves an SSIM of 0.9956 and a PSNR of 44.16 dB, which is over 7 dB higher than other methods, indicating the superior visual quality of the watermarked images. Furthermore, across four different affine transformation tests, GResMark maintains an extraction accuracy exceeding 99 %, demonstrating its exceptional resilience to affine transformation. This improvement can be theoretically explained by the global self-attention and positional modeling capabilities of the Swin Transformer, which allow GResMark to reconstruct spatial relationships between tokens even after complex geometric distortions. Additionally, the LeCA and FCA modules jointly strengthen the model's feature representation, making it more invariant to affine transformations.

4.3. Imperceptibility and robustness against combined distortion training

In real-world scenarios, images are often subjected to various attacks. Therefore, a watermarking model should be capable of resisting



Fig. 8. Visual results of watermarked images with combined distortions.

Table 8
Extraction accuracy with combined distortions.

Attacks	Models								
	(Wu et al., 2025)	RtScW	HiDDeN	TSDL	MBRS	FIN	SepMark	WFormer	GResMark
Cropping(0.4)&JPEG(70)	80.94 %	94.52 %	71.37 %	65.47 %	70.32 %	94.14 %	92.33 %	95.11 %	**98.54 %**
Cropping(0.5)&Scaling(0.7)	88.19 %	69.36 %	70.47 %	62.67 %	78.07 %	50.43 %	84.75 %	86.48 %	**98.38 %**
Cropping(0.5)&Rotation(30)	63.21 %	50.82 %	68.12 %	66.26 %	67.64 %	51.66 %	82.23 %	85.33 %	**97.92 %**
Cropping(0.5)&Scaling(0.7)&Rotation(30)	65.56 %	49.89 %	66.35 %	65.40 %	66.04 %	49.35 %	78.58 %	82.51 %	**97.45 %**
Cropping(0.5)&Affine(s = (0,0.2,0.7,30))	56.23 %	60.85 %	54.34 %	50.66 %	60.94 %	51.33 %	67.22 %	77.14 %	**95.57 %**
Rotation(60)&Affine(s = (0,0.2,0.7,30))	54.72 %	69.54 %	50.48 %	52.71 %	53.62 %	50.11 %	61.45 %	78.02 %	**96.68 %**

multiple types of noise rather than a single specific attack. To achieve this, we train our GResMark with a composite noise layer that incorporates various types of noise. Specifically, the noise layer consists of Identity, JPEG ($QF = 60$), Gaussian Noise ($\sigma = 0.04$), Gaussian Blur ($\sigma = 2$), Median Blur ($W = 5$), Salt-and-pepper Noise ($\sigma = 0.04$), Scaling ($r \in (0.7, 1)$), Cropping ($r \in (0, 0.4)$), Rotation ($\theta \in (-180°, 180°)$), and Affine Transformation ($s = ((-10°, 10°), (0.1, 0.1), (0.8, 0.8), (-30°, 30°))$).

Here, Identity represents the original image. The JPEG attack simulates the process of image compression, which applies block-based discrete cosine transform (DCT) and quantization to remove high-frequency information. Since real JPEG compression is non-differentiable, we adopt a differentiable JPEG simulation layer following MBRS (Jia et al., 2021) to emulate the compression process during training. Gaussian Noise refers to additive noise sampled from a Gaussian distribution. Gaussian Blur applies a Gaussian kernel to scan and modify each pixel in the image. Median Blur replaces each pixel with the median value of its neighboring pixels within a kernel. Salt-and-pepper Noise randomly replaces a proportion ($p \in (0, 1)$) of the pixels in the watermarked image with either black or white pixels with equal probability. Cropping removes part of the image containing the watermark and fills the cropped area with zeros.

During combined attack training, each training batch randomly applies one type of distortion selected from a predefined set of both geometric and non-geometric distortions. To further evaluate the generalization capability of the models, we tested them using parameters beyond those in the training set (e.g., Gaussian Noise ($\sigma = 0.05$), Gaussian Blur ($\sigma = 8$)). To evaluate imperceptibility, we measure the PSNR and SSIM of different models, as reported in Table 6. Our GResMark achieves superior visual quality with a PSNR above 36 and an SSIM exceeding 0.98, both outperforming all compared methods.

Evaluation under Single Distortions: Extraction accuracy for single distortions is shown in Table 7. While GResMark does not achieve SOTA performance in Gaussian Noise and JPEG, it still exhibits notable imperceptibility and robustness. Specifically, the Swin Transformer effectively captures long-range dependencies, but its window-based attention relies on local feature modeling. Gaussian noise introduces random pixel variations that interfere with this local modeling, slightly reducing performance compared to WFormer (Luo et al., 2024) and FIN (Fang et al., 2023c). Similarly, the marginally lower accuracy under JPEG compression arises from the slight misalignment between GResMark's shifted-window self-attention mechanism and JPEG's fixed DCT grid. Nevertheless, the performance gap remains minimal, confirming GResMark's overall robustness. Moreover, when subjected to various attacks, our GResMark demonstrates superior robustness, particularly against geometric distortions. For instance, under rotation attacks, it achieves an extraction accuracy of 100 %, exceeding other methods by more than 10 %.

Evaluation under Combined Distortions: In practical transmission scenarios, watermarked images may undergo various distortions. Therefore, it is crucial to evaluate robustness against combined distortions. To this end, we simulate a set of combined distortions using Cropping, JPEG, Rotation, and Scaling. Fig. 8 provides examples of the watermarked and noise images. The extraction accuracy under these combined attacks is reported in Table 8. HiDDeN (Zhu et al., 2018), TSDL (Liu et al., 2019), and MBRS (Jia et al., 2021) suffer from a significant drop in extraction accuracy under combined attacks, indicating poor robustness. FIN (Fang et al., 2023c) shows relatively strong robustness against Cropping(0.4)&JPEG(70), but fails to address the desynchronization problem caused by geometric distortions, leading to inferior performance under other combined attacks. SepMark (Wu et al., 2023) and WFormer (Luo et al., 2024) show moderate robustness but still fall short under complex attack combinations. In contrast, GResMark demonstrates superior robustness, achieving an extraction accuracy of over 95 % under all combined distortions.

4.4. Imperceptibility and robustness against unseen distortions

In real-world scenarios, a practical watermarking system must be robust to unseen image distortions, rather than relying solely on the specific perturbations encountered during training. To this end, we train models using a limited set of common distortions: Gaussian Blur ($\sigma = 2$), Salt-and-Pepper Noise ($\sigma = 0.04$), Gaussian Noise ($\sigma = 0.04$), and Median Blur ($w = 5$). To assess the model's generalization ability against unseen perturbations, we evaluate it under a diverse set of unseen distortions, including Dropout ($p = 0.7$), Cropout ($p = 0.7$), Cropping ($p = 0.7$), Rotation ($\theta = 5°$), Scaling ($r = 0.9$), Brightness ($f = 3$), and Contrast ($f = 4$). We compared our GResMark with SOTA learning-based methods, including MBRS (Jia et al., 2021), FIN (Fang et al., 2023c), SepMark (Wu et al., 2023), and WFormer (Luo et al., 2024). The experimental results are summarized in Table 9.

As shown in Table Table 9, FIN (Fang et al., 2023c) exhibits low robustness under Dropout, Cropout, and Cropping, while MBRS (Jia et al., 2021), SepMark (Wu et al., 2023), and WFormer (Luo et al., 2024) show improved but still unstable robustness, especially under geometric distortions such as rotation and scaling. In contrast, GResMark consistently

Table 9
PSNR and extraction accuracy under unseen distortions.

Models	PSNR (dB)	ACC (%)						
		Dropout(0.7)	Cropout(0.7)	Cropping(0.7)	Rotation(5°)	Scaling(0.9)	Brightness(3)	Contrast(4)
MBRS	32.88	98.86	90.98	89.13	49.16	48.44	92.25	95.83
FIN	33.41	81.63	77.43	78.14	49.43	50.31	90.01	89.03
SepMark	33.27	95.77	95.82	97.85	50.75	48.02	95.04	95.76
WFormer	34.09	97.67	96.62	97.57	60.28	70.98	97.05	96.70
GResMark	**34.53**	**99.54**	**97.91**	**98.75**	**83.51**	**80.57**	**97.47**	**97.94**

Table 10
Imperceptible and robust performance with different watermark capacities.

Models	Watermark Length	Cropout		S&P		Gaussian Blur		Affine Transformation	
		PSNR(dB)↑	ACC(%)↑	PSNR(dB)↑	ACC(%)↑	PSNR(dB)↑	ACC(%)↑	PSNR(dB)↑	ACC(%)↑
HiDDeN (Zhu et al., 2018)	64	40.62	61.67	46.04	90.42	46.21	84.37	33.58	70.03
	256	43.78	51.84	46.55	54.17	47.85	52.65	35.74	48.67
TSDL (Liu et al., 2019)	64	47.48	93.21	51.16	91.46	45.97	93.21	33.14	68.16
	256	32.54	82.65	30.78	90.44	28.83	86.43	40.55	50.12
MBRS (Jia et al., 2021)	64	48.05	83.50	51.79	96.68	47.91	87.80	35.01	88.39
	256	30.93	79.40	32.28	93.76	30.48	82.56	38.90	53.01
FIN (Fang et al., 2023c)	64	50.61	93.82	52.96	99.98	48.09	98.16	31.69	72.29
	256	48.37	93.34	47.29	99.83	45.56	97.47	26.95	66.64
SepMark (Wu et al., 2023)	64	48.60	93.37	51.86	97.87	48.39	97.97	35.61	90.25
	256	30.38	80.82	36.53	96.39	31.43	90.66	40.34	50.63
WFormer (Luo et al., 2024)	64	52.72	98.35	52.71	99.22	49.36	98.69	36.91	91.79
	256	48.67	96.48	46.89	98.94	45.57	97.84	31.36	86.32
GResMark	64	**54.74**	**99.23**	**55.76**	**99.98**	**52.45**	**100**	**44.16**	**99.08**
	256	**50.20**	**98.21**	**49.62**	**99.99**	**49.48**	**100**	**40.79**	**98.87**

achieves the highest ACC across all unseen distortions, while also maintaining the best image quality (PSNR = 34.53 dB). These results demonstrate GResMark's strong robustness to unseen perturbations and superior overall reliability.

4.5. Watermarking capacity

Similar to imperceptibility and robustness, watermark capacity is a crucial aspect of image watermarking tasks. A higher watermark capacity enables embedding more information into the image, such as detailed copyright descriptions or ownership metadata, which is often required in real-world applications. However, increasing watermark capacity can adversely affect both imperceptibility and robustness. As the capacity grows, additional redundant features are introduced during embedding, potentially reducing the visual quality. Moreover, a larger capacity increases the watermark's exposure to removal by noise and other image-processing attacks.

To evaluate the model's performance under different watermark capacities, we set the watermark capacity to 64 bits and 256 bits, respectively. We trained specific networks for different attacks, including four types of distortions: Cropout ($r = 0.4$), Salt & Pepper Noise ($\sigma = 0.04$), Gaussian Blur ($\sigma = 2$), and Affine Transformations ($s = ((0°, 0°), (0.1, 0.1), (0.7, 0.7), (-30°, 30°))$). During testing, Cropout ($r = 0.5$), Salt & Pepper Noise ($\sigma = 0.05$), Gaussian Blur ($\sigma = 2$), and Affine Transformations ($s = ((10°, 10°), (0.1, 0.1), (0.7, 0.7), (-30°, 30°))$) are used to evaluate both imperceptibility and robustness. We compared our GResMark with SOTA learning-based methods, including HiDDeN (Zhu et al., 2018), TSDL (Liu et al., 2019), MBRS (Jia et al., 2021), FIN (Fang et al., 2023c), SepMark (Wu et al., 2023), and WFormer (Luo et al., 2024). The experimental results are summarized in Table 10.

As shown in Table 10, after embedding a 256-bit watermark, ACC of the HiDDeN model drops sharply to approximately 50, indicating that the model completely loses its robustness when embedding high-capacity watermarks. For MBRS, embedding a 256-bit watermark leads to a significant degradation in watermarked image quality, with the PSNR dropping to approximately 30, and a loss of robustness against affine transformations. Although FIN and WFormer retain some robustness after embedding a 256-bit watermark, both PSNR and ACC are still significantly affected. In contrast, our GResMark achieves the superior imperceptibility and robustness even when embedding a 256-bit watermark.

Table 11
FLOPs computation for different models.

Model	HiDDeN	MBRS	FIN	SepMark	WFormer	GResMark
Params(M)	0.40	5.80	0.75	6.13	1.72	76.33
FLOPs(G)	3.52	13.36	3.56	2.97	13.83	13.91
Speed(im/s)	60.2	17.4	16.9	18.6	14.0	14.3

4.6. Computational complexity

In this section, we analyze the computational cost of the proposed GResMark model from three perspectives: the number of parameters, FLOPs, and image processing speed. As shown in Table 11, GResMark contains 76.33M parameters, which is significantly higher than those of the other models. This increase is primarily due to the large number of parameters introduced by the Swin Transformer backbone. In terms of computational cost, GResMark requires 13.91G FLOPs, which is comparable to WFormer and slightly higher than other watermarking models such as MBRS and FIN. Regarding inference speed, GResMark processes 14.3 images per second, which is close to WFormer but lower than CNN-based watermarking models. Overall, although GResMark demands more parameters and a higher computational load, it achieves superior robustness and resilience to geometric attacks compared with existing deep learning-based watermarking methods, making this trade-off acceptable.

Table 12
Ablation study for network architecture design and constrained loss.

LeCA	FCA	Constrained Loss	Rotation		Scaling		Affine Transformation		JPEG	
			PSNR(dB)↑	ACC(%)↑	PSNR(dB)↑	ACC(%)↑	PSNR(dB)↑	ACC(%)↑	PSNR(dB)↑	ACC(%)↑
✓	✓	✓	**50.26**	**99.98**	**47.56**	**99.85**	**44.16**	99.08	**45.04**	97.68
×	✓	✓	48.22	99.23	46.07	98.79	41.59	98.15	44.32	96.67
✓	×	✓	50.08	99.41	47.53	98.77	43.80	98.04	44.58	96.56
✓	✓	×	48.78	99.96	46.51	99.47	42.92	**99.10**	44.46	**97.72**
✓	×	×	48.83	99.49	46.88	98.45	42.68	98.06	43.89	96.69
×	✓	×	47.36	99.31	45.43	98.53	40.53	97.86	43.72	96.74
×	×	✓	48.69	99.04	46.62	97.69	41.47	96.85	43.75	95.41
×	×	×	47.07	98.92	45.40	97.36	40.07	96.84	43.13	95.50

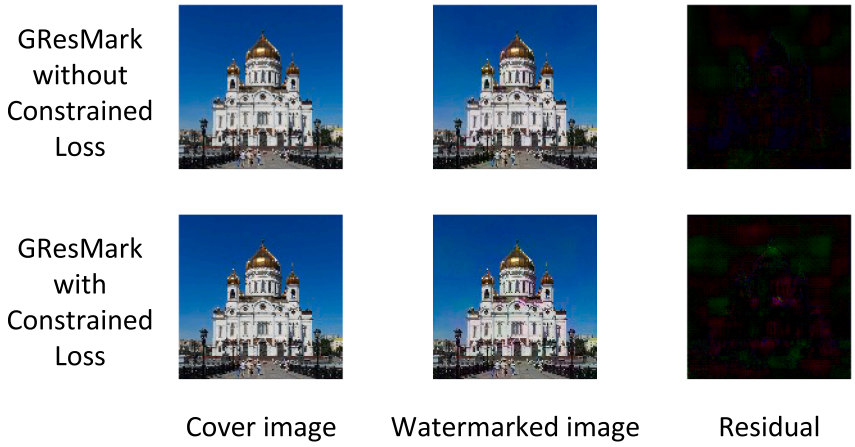


Fig. 9. Visual results of GResMark with or without the Constrained Loss. Residuals represent the difference between the watermarked and cover image and are magnified by a factor of 5 to enhance visibility.

4.7. Ablation study

Effectiveness of LeCA: The LeCA helps GResMark effectively extract both local features and channel information from images. We compared the PSNR and ACC with and without the LeCA, as shown in Table 12. GResMark achieved improvements of 2 dB in PSNR and 0.8 % in ACC. This improvement can be attributed to LeCA's ability to help the Swin Transformer capture both local and channel-wise information. The results highlight the crucial role of LeCA in enhancing GResMark's performance.

Effectiveness of FCA: The FCA helps GResMark learn rich frequency-domain features of images. We compared the PSNR and ACC with and without the FCA, as shown in Table 12. Although the addition of the FCA led to a slight increase in PSNR (by 0.3 dB), it is noteworthy that the ACC was improved (by 0.7 %). This improvement likely stems from FCA's ability to capture rich frequency components, embedding the watermark in a more suitable high-frequency domain.

Effectiveness of Constrained Loss: As illustrated in Fig. 9, the watermarked images generated by GResMark with the Constrained Loss exhibit fewer visual artifacts compared to those without it. Furthermore, as reported in Table 12, the constrained loss improves the imperceptibility of GResMark, with PSNR increasing by approximately 1 dB under various image processing attacks. This improvement is likely because the constrained loss encourages the encoder E to restrict pixel values within the $[0, 255]$ range.

Loss Weight Analysis: As shown in Table 13, when $\lambda_3 = 0.001$ and $\lambda_2 = 10$ are fixed, decreasing λ_1 from 10 to 4 slightly reduces PSNR but significantly improves ACC, indicating stronger watermark robustness with acceptable visual quality. Further reducing λ_1 to 1 causes a large PSNR drop with negligible ACC improvement. Similarly, when adjusting λ_3 from 0.001 to 0.0001, PSNR decreases slightly from 36.15 dB to 35.44 dB, and ACC drops marginally from 99.06 % to 99.01 %. Therefore, we set $\lambda_1 = 4$, $\lambda_2 = 10$, and $\lambda_3 = 0.001$, achieving a good trade-off between watermark invisibility and robustness.

Table 13
Analysis for different loss weight.

Criteria		PSNR(dB)	ACC(%)
λ_1,λ_2,λ_3	10,10,0.001	38.06	96.74
	4,10,0.001	36.15	99.06
	1,10,0.001	33.28	99.42
	4,10,0.0001	35.44	99.01

5. Conclusion

In this article, we propose a geometric attack-resilient image watermarking framework (GResMark) based on Swin Transformer. GResMark leverages the ability of Swin Transformer for global modeling to solve desynchronization caused by geometric distortions. In addition, we design the LeCA to help Swin Transformer capture local and channel information, and introduce the FCA to capture frequency domain features, thereby enhancing GResMark's resilience against high-frequency noise attacks. To prevent some pixel values from falling outside the standard range, we propose a constrained loss, which improves the image quality of watermarked images. Extensive experiments demonstrated that GResMark is highly robust against geometric attacks, outperforming the SOTAs in terms of robustness, imperceptibility, and watermark capacity. Due to its strong robustness and flexibility, GResMark shows great potential for practical applications such as copyright protection, content authentication, and AIGC content traceability, and it can be easily extended to remote sensing and medical imaging scenarios.

Although our GResMark demonstrates remarkable effectiveness against geometric attacks, its robustness against Gaussian noise and JPEG compression has not yet reached the state of the art. Considering the strong resilience of invertible neural networks (INNs) to such distortions, future work will focus on designing a hybrid framework that integrates Transformer modules with INNs, aiming to achieve robustness against both geometric and non-geometric attacks. In addition, the resistance to unseen distortions that are not explicitly modeled in the noise layer during training remains a challenging and unsolved problem. Future research will explore a novel watermarking framework towards better generalization under unseen distortions.

Credit Author Statement

Weitong Chen: Conceptualization, Methodology, Writing – Original Draft, Funding acquisition; **Yuheng Li:** Software, Validation, Writing – Original Draft;**Jiale Zhang:** Supervision, Funding acquisition; **Chunpeng Ge:** Writing – Review & Editing, Funding acquisition. **Di Wu:** Software, Validation ;**Willy Susilo:** Writing – Review & Editing **Shivakumara Palaiahnakote:** Writing – Review & Editing.

Data availability

The data used are all open-source and cited in the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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